prime239v1's L(1, χ) proxy stays extreme, but only through primes already seen

Study `research/nist_seed_l1chi_20261030`, run 2026-10-30 UTC. The run followed the frozen protocol `PROTOCOL.md` / `protocol.json` (sha256 `039665e8…`), merged in PR #2763 before any target was evaluated. The checkpoint gate passed before each run, and the records are in `runs/`.

Consolidated. An independent study, `research/nist_seed_lvalue_20261030` (x = 10^8, exact D_K; canonical), gives the same result. On the 61 shared curves, log L at 10^6 agrees to $2.9 \cdot 10^{-14}$, and prime239v1 is at the 0.247th percentile there. See `CROSSCHECK.md`.

Measured.

- Positive control passes. For secp256k1, $\log P_{\mathrm{DK}} = -0.503074$ against the exact $\log\left(\pi/\left(3\sqrt{3}\right)\right) = -0.503189$, within the calibrated tolerance $\tau = 8.65 \cdot 10^{-4}$. The calibration used 2,000 fundamental discriminants in $[-10^{13}, -10^{12}]$, h from PARI `quadclassunit` (GRH), with mean error $-2 \cdot 10^{-6}$.
- Decisive result. prime239v1 has $\log P_\Delta = -1.4020$ over all primes $\ell \le 10^6$. That is at the 0.246th percentile of the prime-order null, below the 1% threshold, so the frozen verdict is `extreme_persists`. The author's recorded prediction, `not_extreme`, failed. It is the lowest of the 69 eligible curves; next come two unpublished curves (1.31%, 1.53%) and published P-384 (1.69%).

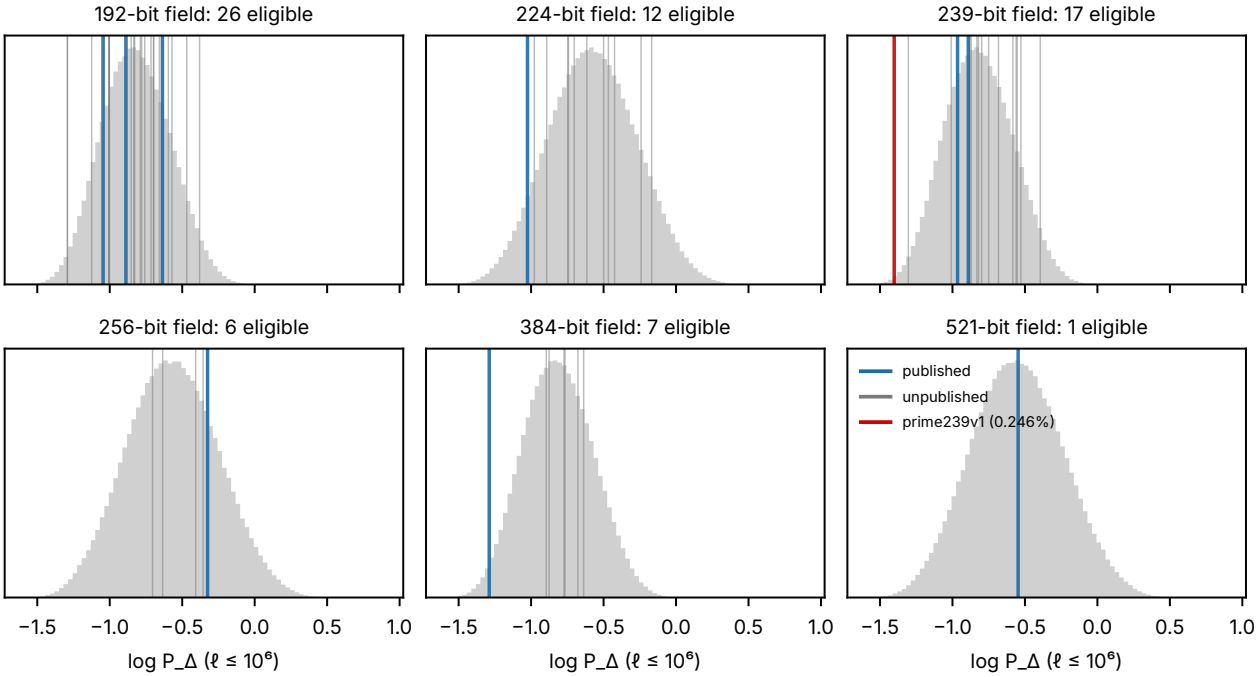


Figure 1: Null density of log P_Δ per field (grey), drawn with the frozen null and seed, with each eligible curve marked. Source: `runs/results.json`, `figures.py`.

Post-hoc, descriptive, not preregistered.

- Where the extreme comes from. The decisive scope was the never-seen terms $5003 < \ell \le 10^6$. They are ordinary: z = −0.26, about the 40th percentile. All terms above $\ell = 1000$ sit near the 72nd percentile. The extreme comes entirely from small primes already seen in the split-count study. The run confirms that signal; it adds no new evidence for it.
- My prediction failed because it extrapolated the unweighted split-count percentile (11% at B = 5003). The $1/\ell$-weighted product is dominated by small ℓ.
- Selection. prime239v1 was chosen for being the minimum of 61 curves seen earlier. Among 69 null draws, a minimum at or below 0.246% has probability $1 - (1 - 0.00246)^{69} \approx 0.16$.
- Class number. $D_K = \Delta$: the conductor is 1, the endomorphism ring is maximal, and $|D_K| \approx 2^{241}$. With $L(1, \chi) \approx 0.246$, $h \approx \sqrt{|D_K|}L/\pi \approx 2^{116.8}$, against about $2^{118.8}$ at $L = 1$. That is about 4 times smaller than typical and astronomically far from the tiny class numbers a CM-method generator produces.
- Published versus unpublished (in-sample). Unpublished curves exceed published ones in 0.699 of 133 within-field pairs (one-sided permutation p = 0.045). The 59 unpublished percentiles fit uniform: KS D = 0.118 against a 5% critical value of 0.177.

Interpretation. The frozen rule reopens only the CM-field question, and the class-number estimate answers it at once: prime239v1's CM field is not small in any generation-relevant sense. No known ECDLP attack depends on L(1, χ) or h at this scale. The selection-rule problem stays open; no hypothesis status changes.